25901211 MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER - Discipline Specific Core Course (DSC)-MG4DSCECO200

MACROECONOMICS-1 (2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum Marks: 70

Part A - Multiple Choice Questions - Answer all questions - Each question carries 1 mark. [1x10-10]

1. c) Real and monetary variables
2. d). Increase production
3. c). Laissez-faire
4. a. J.M Keynes & b) R.F Kahn
5. a). Y = C + I + G+(X-M)
6. c). Government
7. a). Investment
8. c). JM Keynes
9. b). The United Kingdom
10. b). Inversely with the price level

Part B - Short Answer Questions - Answer any 10 questions - Each question carries 3 marks [3x10=30]

11. Discuss the mechanism through which the interest rate ensures that any leakage from the income stream (savings) is returned as investment.

According to Classical economists, the interest rate acts as an automatic equilibrating mechanism between savings and investment. When households save, they withdraw money from the income stream, creating a leakage. This increased supply of loanable funds drives the interest rate downward, which in turn makes borrowing cheaper for firms. Encouraged by lower borrowing costs, entrepreneurs increase investment, thereby re-injecting the leaked savings back into the income stream. Classical economists thus believed that through this flexible interest rate mechanism, savings and investment are always brought into equality without any need for government intervention.

- *1 mark — Identifying savings as leakage and interest rate as the equilibrating mechanism*
- *1 mark — Explaining how lower interest rates encourage investment*
- *1 mark — Concluding that savings and investment are brought to equality automatically*

12. Explain the role of aggregate demand in determining employment according to Keynes.

Keynes fundamentally rejected the Classical notion that supply creates its own demand (Say's Law). He argued that the level of employment in an economy is determined by the level of effective demand, AD = C + I) Entrepreneurs decide how much to produce and how many workers to employ based on their expectations of aggregate demand. If aggregate demand is insufficient, output contracts and involuntary unemployment emerges. Keynes concluded that full employment is not automatic but can only be achieved when aggregate demand is raised to an adequate level, which may require deliberate government intervention.

- *1 mark — Rejecting Say's Law and introducing effective demand (AD = C + I)*
- *1 mark — Explaining how insufficient AD leads to involuntary unemployment*
- *1 mark — Concluding that full employment requires deliberate raising of aggregate demand*

13. How does Fisher explain the relationship between money supply and price level?

Irving Fisher explained this relationship through his famous **Equation of Exchange: MV = PT**, where M is the money supply, V is the velocity of circulation, P is the price level, and T is the volume of transactions. Fisher assumed that V and T remain constant in the short run. Under these assumptions, any increase in the money supply leads to a direct and proportional increase in the price level. Doubling M will double P. Money is therefore considered neutral — it affects only nominal variables like prices and not real variables like output or employment.

- *1 mark — Stating the equation MV = PT with explanation of terms*
- *1 mark — Explaining the assumptions of constant V and T*
- *1 mark — Concluding that money supply and price level are directly and proportionally related*

14. Identify the main postulates of Classical theory of Prices.

The Classical Theory of Prices rests on a set of interrelated postulates. They can be listed as follows - Quantity Theory of Money — Neutrality of Money - Flexible Prices and Wages - Full Employment - Say's Law of Market - Long-run focus

- *1/2 mark each for postulates correctly identified and explained*

15. Define the concept of effective demand in Keynesian economics

This concept is central to the entire Keynesian system. Effective demand refers to the level of aggregate demand at which the economy reaches equilibrium, that is, where aggregate supply equals aggregate demand. It represents the point at which entrepreneurs' expected revenues match their costs of production, and it is this expectation that determines how much output they will produce and how many workers they will employ. A deficiency of effective demand results in unemployment, while excess demand leads to inflation.

- *1 mark — Defining effective demand as the equilibrium point of AS and AD*
- *1 mark — Explaining that it determines output and employment decisions of entrepreneurs*
- *1 mark — Stating that deficiency leads to unemployment and excess leads to inflation*

16. Why do classical economists oppose fiscal and monetary policy intervention?

Classical economists oppose government intervention on the following grounds: Say's Law - Flexible wages and prices - Crowding out effect - Quantity Theory - They believe intervention by authorities distorts market signals and leads to inefficiency - The economy is assumed to be inherently stable in the long run

- *1 mark each for any three grounds correctly identified and briefly explained*

17. What is meant by Permanent Income?

The concept of Permanent Income was introduced by Milton Friedman as the foundation of his Permanent Income Hypothesis. Permanent income refers to the expected long-run average income that an individual anticipates receiving steadily over their lifetime, based on their assets, skills, and earning capacity. Friedman distinguished this from transitory income, which represents unexpected or temporary gains and losses. He argued that people base their consumption decisions on permanent income rather than current income, smoothing out their spending over time. A temporary windfall, being transitory, would largely be saved rather than consumed, as it does not alter the individual's perception of their long-run income level.

- *1 mark — Defining permanent income as expected long-run average income*
- *1 mark — Distinguishing it from transitory income*
- *1 mark — Explaining that consumption is based on permanent income, not current income*

18. Examine the factors that dampens accelerator

The accelerator principle states that investment is induced by changes in output/income. However, several factors dampen its effect: Excess capacity - Expectations - Capital indivisibility - Time lags - Financial constraints - Availability of labour

- *1 mark each for any three factors correctly identified and briefly explained*

19. How does stagflation differ from regular inflation?

Regular inflation refers to a sustained rise in the general price level, typically associated with rising output and low unemployment, often caused by excess demand in the economy. Stagflation is a situation where high inflation coexists with economic stagnation and rising unemployment. Stagflation is generally caused by supply-side shocks, as seen during the 1970s oil crisis, rather than excess demand, which is why conventional demand management tools prove inadequate.

- *1 mark — Defining regular inflation with its causes and characteristics*
- *1 mark — Defining stagflation and its coexistence of inflation and unemployment*
- *1 mark — Explaining the difference in causes and why conventional policy tools fail*

20. Compare Ricardian Equivalence with the Keynesian view on fiscal multipliers.

The Keynesian view holds that an increase in government spending raises aggregate demand directly, and through the multiplier effect, and increases income and employment. The fiscal policy is very effective and the fiscal multiplier is greater than one.

But Ricardian Equivalence, associated with Robert Barro building on Ricardo, challenges this view entirely. It argues that rational, forward-looking consumers, upon observing a rise in government borrowing, anticipate higher taxes in the future to repay the debt. They therefore increase their savings today to prepare for that future burden, fully offsetting the government's interventions. The result is a fiscal multiplier of zero, rendering fiscal policy ineffective.

- *1 mark — Explaining the Keynesian view with multiplier greater than one*
- *1 mark — Explaining Ricardian Equivalence and the anticipation of future taxes*
- *1 mark — Contrasting the two — multiplier greater than one versus multiplier of zero*

21. Define the term 'k' in the Cambridge approach.

In the Cambridge Cash Balance approach developed by Marshall and Pigou, the money demand equation is written as $M = kPY$, where M is the money supply, P is the price level, and Y is real income. The term 'k', known as the Cambridge k, represents the fraction of nominal income that individuals choose to hold in the form of cash or money balances rather than spending it.

- *1 mark — Stating the equation $M = kPY$ with explanation of terms*
- *1 mark — Defining k as the fraction of nominal income held as cash balances*
- *1 mark — Explaining its behavioural significance or its relationship to velocity*

22. Define marginal propensity to consume (MPC). Why is it significant?

MPC is the ratio of the change in consumption to a change in income: $MPC = \Delta C / \Delta Y$. It measures how much of an additional unit of income is spent on consumption. According to Keynes's Psychological Law MPC lies between 0 and 1. MPC determines the size of the fiscal multiplier -

Higher MPC → larger multiplier → greater impact of government spending, MPC is central to understanding consumption function and aggregate demand and helps in predicting the economy's response to income changes and also explains the turning points of the trade cycle.

- *1 mark — Defining MPC with the formula ΔC/ΔY and the range between 0 and 1*
- *1 mark — Explaining its role in determining the multiplier*
- *1 mark — Stating its significance in understanding consumption function, aggregate demand, and trade cycle turning points*

23. Distinguish between open inflation and suppressed inflation.

Open inflation refers to a situation where the general price level rises freely and visibly as a result of excess demand over supply, without any government interference. Prices are allowed to adjust upward through normal market forces, making the inflation observable and measurable.

Suppressed inflation, also called repressed inflation, occurs when inflationary pressure exists in the economy but is held back artificially through government controls such as price ceilings, rationing, and subsidies. While prices appear stable on the surface, the underlying imbalance between demand and supply persists.

- *1 mark — Defining open inflation with its key characteristic of freely rising prices*
- *1 mark — Defining suppressed inflation and the role of government controls.*
- *1 mark — Identifying the key distinction*

<u>Part C - Short Essay Questions - Answer any 5 questions - Each question carries 6 marks</u>

24. What mechanism restores equilibrium if the supply of labor exceeds the demand?

Classical economists believed the labor market can fix itself without any outside help— if there are more workers looking for jobs than there are jobs available, competition among workers will naturally push wages down. This is the market's built-in correction mechanism. employers will hire more workers. At the same time, some workers who feel the lower wage is simply not worth their time will leave the market. So demand for labor goes up while supply comes down — and the gap between them gradually closes until the market finds a new balance. At this new equilibrium, everyone who genuinely wants to work at the going wage has a job. If some people remain unemployed, Classical economists called it voluntary unemployment.

- *2 marks — Explaining that excess labour supply pushes wages down through competition among workers*
- *2 marks — Explaining the dual adjustment — employers hire more as labour becomes cheaper, and some workers voluntarily exit the market*
- *2 marks — Concluding that the market reaches a new equilibrium and any remaining unemployment is voluntary*

25. Explain the role of wage and price flexibility in the classical macroeconomic model. Why is it essential for achieving full employment equilibrium?

The Classical model runs on one big assumption — that wages and prices can move freely in any direction, up or down, without any friction. In the labor market, if there are too many workers chasing too few jobs, wages fall. Falling wages make hiring cheaper for employers and discourage some workers from participating, and these two forces together bring the market back into balance. The goods market works the same way — if goods are piling up unsold, prices fall, buyers are attracted

back, and the excess clears itself. Both markets are essentially self-cleaning, and their interaction ensures the whole economy stays close to full employment. Flexible wages and prices as the economy's natural shock absorbers which will bring back the economy to equilibrium. (Keynes questioned the whole thing).

- *2 marks — Explaining wage flexibility in the labour market and how it restores equilibrium*
- *2 marks — Explaining price flexibility in the goods market and how it clear excess supply*
- *2 marks — Explaining why flexibility is essential as the economy's tool for equilibrium*

26. Evaluate the effectiveness of Keynesian Fiscal policy as a response to severe economic crisis

Keynesian fiscal policy is essentially the government stepping in to spend and adjust taxes when the economy is in trouble. Keynes argued that when private spending collapses, Government has to fill the gap, The mechanism works through the multiplier. Government spending puts money in people's hands, they spend a portion of it, that spending becomes someone else's income, and so the cycle continues and boost the economy. Tax cuts will leave more money in people's pockets to spend. We have examples from which supports this principle like The New Deal, post-war reconstruction, and the 2008 stimulus packages all showed that well-timed government spending can pull economies out of deep downturns.

- 2 marks — Explaining the rationale for fiscal intervention
- 2 marks — Explaining the multiplier mechanism through which government spending and tax cuts
- 2 marks — Evaluating with historical evidence (New Deal, post-war reconstruction, 2008 stimulus) and briefly acknowledging limitations such as crowding out and time lags

27 Government intervention makes the Keynesian model more realistic - Examine this statement with reference to the three-sector model.

The simple Keynesian model includes only households and firms, making it an incomplete picture of modern economies. The three-sector model corrects this by introducing government spending (G) as an injection and taxation (T) as a leakage, with the new equilibrium condition $Y = C + I + G$ and leakages-injections balance $S + T = I + G$. This makes the model far more useful. It explains how government spending raises income through the multiplier, how taxation cools excess demand, and how even a balanced budget produces a net increase in income through the balanced budget multiplier. It explains the government's stabilising role. Thus, with government Keynesian model becomes a practical tool for managing real economies.

- *2 marks — Explaining that the two-sector model is incomplete and how the three-sector model introduces G as an injection and T as a leakage*
- *2 marks — Explaining the effects of government spending and taxation*
- *2 marks — Explaining the government's stabilising role during downturns and booms, and concluding*

28. Explain the MEC Schedule and why it slopes downward

The Marginal Efficiency of Capital (MEC) is the expected rate of return on an investment project. An entrepreneur compares this expected return against the prevailing interest rate — if MEC exceeds the interest rate, the investment is worth making; if it falls short, the project is dropped. Investment continues until MEC and the interest rate are equal.

The MEC Schedule plots how MEC changes as the volume of investment increases, and it slopes downward for two straightforward reasons. First, as investment rises, the demand for capital goods

pushes up their prices, making each additional unit of investment more expensive and therefore less profitable. Second, and more fundamentally, diminishing returns set in — each additional unit of capital adds less to output than the one before it, so expected returns progressively decline. Entrepreneurs naturally chase the most profitable projects first and work their way down to less attractive ones, pulling the MEC lower with each step. (Credit should be given to explanation with diagram)

- *2 marks — Defining MEC clearly and explaining the investment decision rule*
- *2 marks — Explaining the first reason for downward slope — rising supply price of capital goods as investment increases*
- *2 marks — Explaining the second reason — diminishing marginal returns causing expected yields to fall progressively, with credit for a relevant diagram*

29 Evaluate the difference between Short run and Long run consumption function with the help of diagrams.

The short-run consumption function, based on Keynes, shows consumption as **C = a + bY**, where there is positive autonomous consumption and the curve starts above the origin with a declining APC. In contrast, the long-run consumption function, supported by Kuznets and later theories, is proportional and passes through the origin, indicating constant APC. In the short run, consumption depends mainly on current income, while in the long run it reflects broader factors like permanent or lifetime income. Graphically, the short-run curve is upward sloping with an intercept, whereas the long-run curve is a straight line from the origin. Thus, the key difference lies in the presence of autonomous consumption and the stability of consumption behaviour over time. (Credit should be given to explanation with diagram)

- *2 marks — Explaining the short run consumption function*
- *2 marks — Explaining the long run consumption function*
- *2 marks — Distinguish the two with credit for a combined diagram*

30. Examine the role of Fiscal policy during inflation.

Inflation is essentially a situation where too much money chases too few goods. Fiscal policy tackles this problem by withdrawing excess purchasing power from the economy. The most direct method is cutting government expenditure, which reduces aggregate demand and relieves the upward pressure on prices. Alongside this, raising taxes leaves households and firms with lesser money in hand, thereby squeezing demand further. When the government collects more than it spends, it withdraws purchasing power from the economy and exercises a dampening effect on prices. The government may also borrow from the public through bonds and securities, which mops up excess liquidity from the system and reduces the funds available for spending, thus helping to keep inflationary pressures in check.

- *2 marks — Explaining how cutting government expenditure directly reduces aggregate demand*
- *2 marks — Explaining how raising taxes reduces disposable income and squeezes demand,*
- *2 marks — Explaining how government borrowing through bonds and securities clears excess liquidity and concluding that contractionary fiscal policy is an effective instrument for price stability*

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